# SEACOM ENGINEERING COLLEGE

**NAME – BIMAL MAITY**

**ROLL NO – 20600224035**

**REG NO – 242060120318**

**SUBJECT- DBMS**

**YEAR - 3rd**

**SEM – 6th**

**DEPT- INFORMATION TECHNOLOGY**

**TOPIC- Entity–Relationship (ER) Model**

# 1. Entity–Relationship (ER) Model

The **Entity–Relationship (ER) Model** is a high-level conceptual data model used in DBMS to design and represent the structure of a database.

It was introduced by **Peter Chen** in 1976.

It visually represents data using **entities, attributes, and relationships**.

## Components of ER Model

### 1. Entity

An **entity** is a real-world object or concept that can be identified uniquely.

Examples:

- Student
- Employee
- Department

**Types of Entities:**

1. **Strong Entity**
   - Has its own primary key.
   - Represented by a single rectangle.
2. **Weak Entity**
   - Does not have its own primary key.
   - Depends on a strong entity.
   - Represented by double rectangle.

## 2. Attribute

An **attribute** describes properties of an entity.

Example:
Student → Roll_No, Name, Age

**Types of Attributes:**

1. **Simple Attribute** – Cannot be divided (Age)
2. **Composite Attribute** – Can be divided (Name → First, Last)
3. **Single-Valued Attribute** – One value (DOB)
4. **Multi-Valued Attribute** – Multiple values (Phone Numbers)
5. **Derived Attribute** – Calculated from another (Age from DOB)
6. **Key Attribute** – Uniquely identifies entity (Roll_No)

## 3. Relationship

A **relationship** represents association between entities.

Example:
Student — Enrolls — Course

Represented by diamond shape.

## Types of Relationships (Cardinality)

1. **One-to-One (1:1)**
   One entity related to one entity.
   Example: Person – Passport
2. **One-to-Many (1:M)**
   One entity related to many entities.
   Example: Department – Employees
3. **Many-to-Many (M:N)**
   Many entities related to many entities.
   Example: Student – Course

## ER Diagram Symbols

| Component | Symbol |
|---|---|
| Entity | Rectangle |
| Weak Entity | Double Rectangle |
| Relationship | Diamond |
| Attribute | Oval |
| Key Attribute | Underlined |
| Multi-valued Attribute | Double Oval |
| Derived Attribute | Dashed Oval |

## Example ER Diagram

Example: Student Database

```
[STUDENT] ---- (ENROLLS) ---- [COURSE]
    |                            |
  Roll_No                      Course_ID
  Name                         Course_Name
  Age
```

## Advantages of ER Model

- Easy to understand
- Provides clear database structure
- Reduces redundancy
- Helps in database design
- Improves communication between users and developers